A linear KL family and the search for webs

Consolidated research note

6 November 2026

Abstract

We give explicit permutations in finite symmetric groups with ordinary Kazhdan–Lusztig coefficient $\mu = m + 1$, explain the geometric design, and prove the full polynomial by a Schubert slice and the decomposition theorem. Together with known small examples, this realizes every nonnegative integer as an ordinary finite-type μ-coefficient. We give an exact finite Schur realization, compare the family with the finite exponential and affine row-sum constructions, and explain the additional problem of finding a fixed-rank $\mathfrak{sl}_3$ web model. A conditional bending construction transports the coefficient to four-row rectangular canonical labels. We give its label algorithm, separate the required canonical-basis inputs, and display six hourglass growth representatives. The previously reported additive three-colour calculation has coefficient $2k$; its canonical completion remains unresolved.

1 The linear family and where it comes from

We use ordinary Kazhdan–Lusztig polynomials $P_{x,y}(q)$. For $x < y$ in Bruhat order, $\mu(x, y)$ is the coefficient at $q^{(\ell(y)-\ell(x)-1)/2}$ when the length difference is odd, and is zero when the difference is even. Thus the exponent is the maximum permitted degree, whether or not the polynomial attains that degree. Set $[r]_q = 1 + q + \cdots + q^{r-1}$.

Theorem 1. *For every integer $m \geq 3$, put $h = 2m + 3$ and $N = 3m + 3$. In one-line notation, define*

$$x_m = (3, 2, 1,\ h, h-1, \ldots, 4,\ N, N-1, \ldots, h+1), \tag{1}$$

$$y_m = (h, 3, 2, N,\ h-2, h-1, h-4, h-3, \ldots, 5, 6,\ 1,\ N-1, N-2, \ldots, h+1,\ 4). \tag{2}$$

The middle string of y_m consists of $m-1$ ascending adjacent pairs, ordered from largest to smallest. Then $x_m < y_m$ in S_N, and

$$\boxed{P_{x_m,y_m}(q) = [3]_q[m]_q(1+q)^m - q^{2m} - q^{2m+1}.} \tag{3}$$

Moreover,

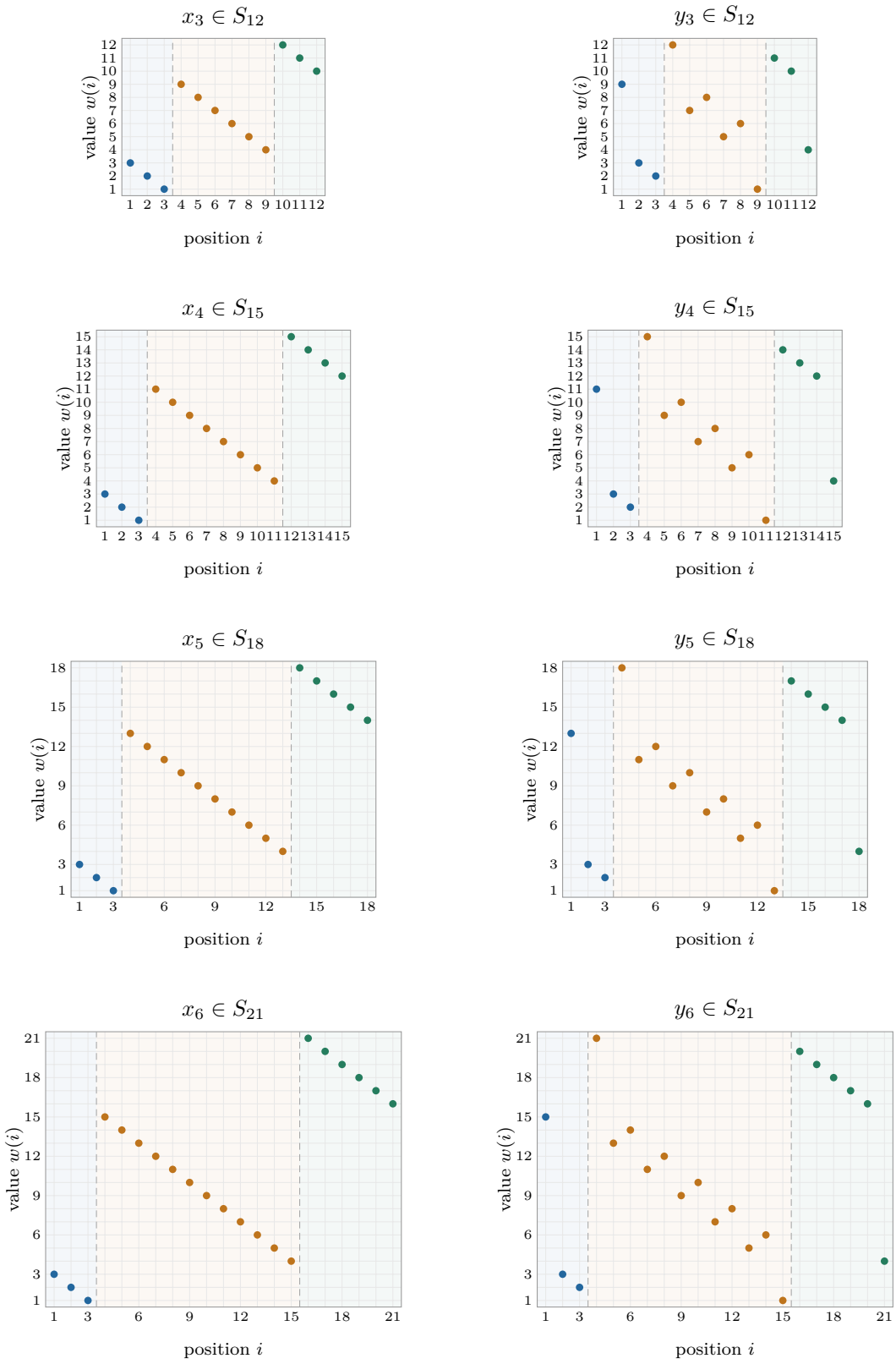
$$\ell(x_m) = \frac{5m^2 - 3m + 6}{2}, \qquad \ell(y_m) = \frac{5m^2 + 5m + 8}{2},$$

so

$$\boxed{\mu(x_m, y_m) = m + 1 = N/3.}$$

Figures 1 and 2 show the family in dot and string notation. It realizes $4, 5, 6, \ldots$ consecutively. For example,

m	3	4	5	6
N	12	15	18	21
$\ell(x_m)$	21	37	58	84
$\ell(y_m)$	34	54	79	109
$\mu(x_m, y_m)$	4	5	6	7



Source-position blocks: 3 $2m$ m.

Figure 1: The first four pairs, in $S_{12}, S_{15}, S_{18}, S_{21}$, with $\mu = 4, 5, 6, 7$. A dot at $(i, w(i))$ records one entry of the permutation. Colours distinguish the three source-position blocks of sizes $3, 2m, m$. The descending middle block of x_m is replaced in y_m by the staircase of ascending pairs and four displaced entries. These are permutation diagrams, not invariant webs.

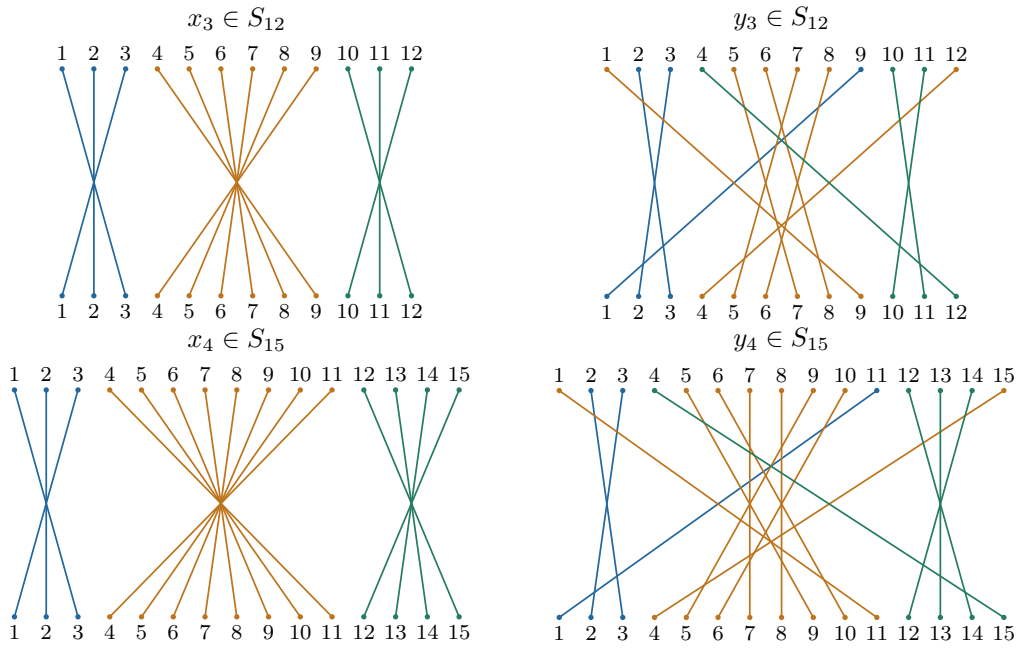


Figure 2: String diagrams for the first two family members: $m = 3$ in S_{12} above and $m = 4$ in S_{15} below. The strand starting at bottom position i ends at top position $w(i)$; its colour records its source-position block of size 3, $2m$ or m. Intersections are crossings, not vertices. The two pairs have $\mu = 4$ and 5, respectively.

Corollary 2. *Every nonnegative integer occurs as an ordinary μ-coefficient for a pair of permutations in some finite symmetric group.*

Proof. For an integer $k \geq 4$, take $m = k - 1$ in Theorem 1; the pair lies in S_{3k}. The values 2 and 3 already occur in S_{12} and S_{11}, respectively, by [15, Theorem 1]. A simple reflection gives $\mu = 1$; the pair $123 < 231$ has even length difference and gives $\mu = 0$. □

1.1 Reverse engineering the geometry

A productive approach is to work backward from the geometric meaning of μ. For a Schubert slice Y of odd dimension $d = \ell(y) - \ell(x)$, with IC_Y normalized to be $\mathbb{Q}[d]$ on its smooth locus, the ordinary polynomial is

$$P_{x,y}(q) = \sum_{j \geq 0} \dim \mathcal{H}^{2j-d}(\mathrm{IC}_Y)_0 \, q^j.$$

Its maximum permitted exponent corresponds precisely to stalk degree -1:

$$\mu(x, y) = \dim \mathcal{H}^{-1}(\mathrm{IC}_Y)_0.$$

Suppose a proper resolution $\pi : \widetilde{Y} \to Y$ is small away from 0, its exceptional fiber Z is smooth and irreducible of dimension D, and $d = 2D - 1$. Proper base change places the unique top class $H^{2D}(Z; \mathbb{Q})$ in stalk degree $+1$ of $R\pi_*\mathbb{Q}[d]$. The decomposition theorem splits off $\mathbb{Q}_0[-1]$, since the middle intersection complex has no stalk cohomology in nonnegative degrees at 0. Self-duality also splits off $\mathbb{Q}_0[1]$. No other nonzero-degree point shifts are possible, by fiber dimension and self-duality. Consequently

$$\dim \mathcal{H}^{-1}(\mathrm{IC}_Y)_0 = \dim H^{2D-2}(Z; \mathbb{Q}) - 1 = b_2(Z) - 1. \tag{4}$$

A possible degree-zero point summand does not affect this calculation.

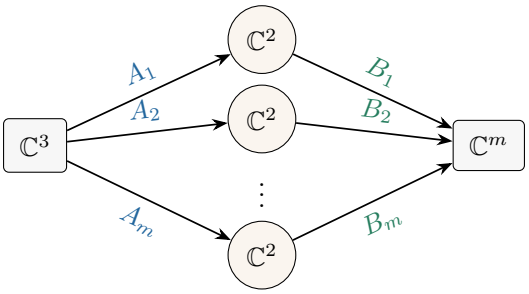
This suggests designing a fiber with an increasing second Betti number. McNamara's model remembers a common kernel hyperplane, intermediate lines, and a common image line. Keep the endpoint dimensions a, b free, and consider m pairs of maps

$$A_i : \mathbb{C}^a \to \mathbb{C}^2, \qquad B_i : \mathbb{C}^2 \to \mathbb{C}^b,$$

with the combined A and B each of rank at most one and with $B_i A_i = 0$. The incidence resolution has exceptional fiber

$$Z = \mathbb{P}^{a-1} \times (\mathbb{P}^1)^m \times \mathbb{P}^{b-1}.$$

For $a, b \geq 2$, each factor contributes one class to H^2, so $b_2(Z) = m + 2$.



$B_i A_i = 0$ for every branch i;
all A_i have a common kernel hyperplane,
and all B_i have a common image line.

Figure 3: The maps for $a = 3$, $b = m$. Each branch has dimension two; the common endpoints have dimensions three and m. The rank-one conditions couple the branches through common endpoint data. This is a quiver of vector spaces and linear maps, not an $\mathfrak{sl}_3$ spider diagram.

The intermediate dimension two is useful. If either A_i or B_i is nonzero, its image or kernel determines the intermediate line. Only when both vanish does a free $\mathbb{P}^1$ occur. After fixing the common endpoint data, a branch is the three-dimensional variety

$$\{(u, \varphi) \in \mathbb{C}^2 \times (\mathbb{C}^2)^* : \varphi(u) = 0\}.$$

Making both maps zero has codimension three and creates a one-dimensional fiber, giving the favorable smallness inequality $3 > 2$.

The resolution is a rank-$2m$ vector bundle over Z. Therefore

$$D = a + b + m - 2, \qquad d = a + b + 3m - 2,$$

and the desired condition $d = 2D - 1$ becomes

$$a + b = m + 3.$$

The remaining smallness margins, along the loci where all A_i or all B_i vanish, are $m + 1 - a$ and $m + 1 - b$. Thus we require

$$a + b = m + 3, \qquad a \leq m, \qquad b \leq m.$$

These conditions force both endpoints to be at least three. The simplest choice with one endpoint fixed is $a = 3$, $b = m$. Choosing $a = 2$ would instead force $b = m + 1$, where strict smallness fails.

The permutation pattern is obtained by translating these equations into essential rank conditions. Reversing blocks of sizes $a, 2m, b$ gives a chart with three variable off-diagonal blocks. A staircase

of rank conditions imposes the cumulative equations $\sum_{i\le s} B_i J_2 A_i = 0$. Successive differences impose the individual zero compositions. The staircase moves two rows upward and two columns rightward, producing the ascending pairs in (2). Section 2 verifies this translation and all required geometric properties.

1.2 Geometric ancestry and comparison

The rank-one, zero-composition model is adapted from McNamara's construction [9, §4], which in turn follows Kashiwara–Saito, Polo, and McNamara–Williamson. The dimensions $(3, 2, \dots, 2, m)$ are an unequal-dimension modification of his template, not a literal specialization of its equal-dimension parameters. The proof below establishes the necessary Schubert identification directly.

Our first member has $\mu = 4$. The supplied note reports a chain of 78 μ-preserving star, symmetry and cancellation moves joins it to Warrington's published Figure 4 pair

$$(3, 2, 1, 7, 6, 5, 4, 10, 9, 8), \qquad (6, 7, 8, 9, 10, 2, 3, 4, 5, 1)$$

in S_{10} [15]. This is μ-equivalence; the full polynomials and their length gaps differ. The chain certificate was not supplied and is not used in the proof below. We have not established that the all-m formula is new, or identified the entire family with a previously published family.

Scott–Xi's family is in extended affine type $\widetilde{A}_n$ and has $\mu = n + 2$ [16, Theorem 3.3]. Its critical degree is n, whereas ours is $2m$. A relation between the two constructions has not been proved. Their shared linear growth alone gives no such identification.

2 Proof of the ordinary polynomial and mu value

We first keep $a, b \ge 2$ and $m \ge 1$ arbitrary. Put $h = a + 2m$ and $N = h + b$, and write

$$\begin{aligned} x &= (a, a-1, \dots, 1,\ h, h-1, \dots, a+1,\ N, N-1, \dots, h+1), \\ y &= (h, a, a-1, \dots, 2,\ N,\ h-2, h-1, \dots, a+2, a+3, \\ &\qquad 1,\ N-1, \dots, h+1,\ a+1). \end{aligned}$$

The paired string is empty when $m = 1$. We will specialize to $a = 3$, $b = m$ after identifying the slice.

The transverse chart and its normalization

Let $G = \mathrm{GL}_N(\mathbb{C})$, let $\mathcal{B}$ be the upper triangular Borel, and put $X_w = \overline{\mathcal{B}w\mathcal{B}/\mathcal{B}}$. Flags are represented by the spans of the initial columns of a matrix. Let J_r denote the antidiagonal identity matrix.

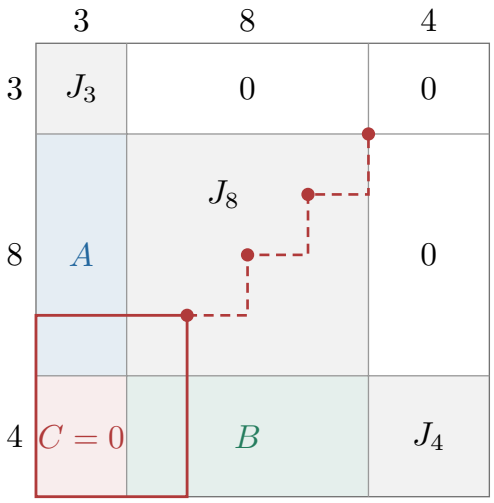
The permutation x is the longest element of $S_a \times S_{2m} \times S_b$. Let U^-_{cross} be the block lower unitriangular group for these three blocks. Factoring the block diagonal lower triangular stabilizer out of $\mathcal{B}^- x\mathcal{B}/\mathcal{B}$ gives unique representatives

$$M = \begin{pmatrix} J_a & 0 & 0 \\ A & J_{2m} & 0 \\ C & B & J_b \end{pmatrix}, \quad A \in \mathrm{Mat}_{2m\times a},\ B \in \mathrm{Mat}_{b\times 2m},\ C \in \mathrm{Mat}_{b\times a}\,. \tag{5}$$

The origin is the flag $x\mathcal{B}$.

For the local product, let U_x be the block diagonal upper unitriangular group. Reversing each block gives a unique algebraic factorization

$$xU^-x^{-1} = U_x U^-_{\mathrm{cross}}.$$



$m = 4, \quad h = 11, \quad N = 15$

Red corners: $(i, j) = (h - 2s, 3 + 2s)$.

$s = 1$ selects the outlined rectangle:

$\begin{pmatrix} A_{\text{last } 2} & J_2 \\ 0 & B_{\text{first } 2} \end{pmatrix}$.

rank ≤ 2 gives $B_1 J_2 A_1 = 0$.

The next corner gives

$B_1 J_2 A_1 + B_2 J_2 A_2 = 0$.

Successive differences separate the branches.

Figure 4: The transverse matrix chart at $m = 4$. Southwest essential rectangles impose rank bounds along the red staircase. Eliminating their antidiagonal pivot blocks yields cumulative zero-composition equations; subtracting consecutive equations separates the branches.

The inverse factorization extracts the diagonal blocks and multiplies by their inverses, so it is regular. Since $U_x \subset \mathcal{B}$ and X_y is $\mathcal{B}$-stable, multiplication identifies the open chart on X_y with

$$U_x \times Y, \qquad Y = X_y \cap \mathcal{B}^- x \mathcal{B}/\mathcal{B}.$$

Here $\dim U_x = \ell(x)$. This identifies the smooth factor explicitly and fixes the shift on the slice.

All rank conditions on the slice

The Schubert rank conditions in this convention are

$$\operatorname{rank} M_{\{i+1,\ldots,N\},\{1,\ldots,j\}} \leq r_y(i, j), \qquad r_y(i, j) = \#\{t \leq j : y(t) > i\}. \tag{6}$$

In southwest orientation, the essential corners satisfy

$$y(j) \leq i < y(j + 1), \qquad y^{-1}(i) \leq j < y^{-1}(i + 1). \tag{7}$$

Fulton's essential-set theorem [11, §3], with rows reversed, says that these conditions imply all of (6). Their orientation also follows from elementary rank propagation: one may enlarge a rectangle without changing its bound, or restore a row or column with a rank increase of at most one. The simultaneous failure of those reductions is exactly (7).

For this y, its positional ascents are $j = a$, $j = h$, and $j = a + 2s$ for $1 \leq s < m$. The ascents of y^{-1} occur at $i = a$, $i = h$, and $i = h - 2s$ for $1 \leq s < m$. At an internal positional ascent, the adjacent values are $h - 2s, h - 2s + 1$, so (7) allows only $i = h - 2s$. At $j = a$ or $j = h$, it allows $i = a, h$ and excludes the internal values. Counting entries above each threshold gives the complete table

i	j	$r_y(i, j)$
a	a	1
h	a	0
h	h	1
$h - 2s$	$a + 2s$	$2s$ $(1 \leq s \leq m)$

where $s = m$ includes the corner (a, h).

The first three rows impose $C = 0$, $\operatorname{rank} A \le 1$, and $\operatorname{rank} B \le 1$. At the remaining corner indexed by s, the southwest rectangle is

$$\begin{pmatrix} A_{\text{last } 2s \text{ rows}} & J_{2s} \\ 0 & B_{\text{first } 2s \text{ columns}} \end{pmatrix}.$$

Eliminating its fixed invertible pivot block shows that its rank is

$$2s + \operatorname{rank}(B_{\text{first } 2s \text{ columns}} J_{2s} A_{\text{last } 2s \text{ rows}}).$$

The bound $2s$ is therefore equivalent to the vanishing of this product.

Number the two-row blocks of A from bottom to top and the two-column blocks of B from left to right. The products are $\sum_{i=1}^{s} B_i J_2 A_i$. Taking successive differences gives $B_i J_2 A_i = 0$ for every i. Absorbing J_2 into each A_i, an invertible linear change preserving the overall rank of A, identifies the slice with

$$Y_{a,b,m} = \left\{ (A_i, B_i)_{i=1}^{m} : \begin{array}{c} A_i : \mathbb{C}^a \to \mathbb{C}^2,\ B_i : \mathbb{C}^2 \to \mathbb{C}^b, \\ \operatorname{rank} \begin{pmatrix} A_1 \\ \vdots \\ A_m \end{pmatrix} \le 1,\ \operatorname{rank}(B_1\ \cdots\ B_m) \le 1, \\ B_i A_i = 0 \text{ for all } i \end{array} \right\}. \tag{8}$$

These equations identify the reduced complex variety, which is sufficient for intersection cohomology. The pivot operations also identify the corresponding determinantal ideals.

The origin satisfies every condition, proving $x \le y$. For the lengths,

$$\ell(x) = \binom{a}{2} + \binom{2m}{2} + \binom{b}{2}.$$

Counting the inversions contributed successively by h, the entries $a, \dots, 2$, the entry N, the ascending pairs, and the final descending block gives

$$\ell(y) = (h-1) + \binom{a}{2} + (2m + b - 1) + 2m(m-1) + \binom{b}{2}.$$

Hence

$$\ell(y) - \ell(x) = a + b + 3m - 2 > 0. \tag{9}$$

A resolution and smallness away from the origin

Let

$$Z = \mathbb{P}((\mathbb{C}^a)^*) \times (\mathbb{P}^1)^m \times \mathbb{P}(\mathbb{C}^b).$$

A point of Z consists of a hyperplane $H \subset \mathbb{C}^a$, lines $L_i \subset \mathbb{C}^2$, and a line $L \subset \mathbb{C}^b$. Form the vector bundle

$$E = \bigoplus_{i=1}^{m} \left(\operatorname{Hom}(\mathbb{C}^a/H, L_i) \oplus \operatorname{Hom}(\mathbb{C}^2/L_i, L) \right). \tag{10}$$

Its total space $\widetilde{Y}$ maps to $Y_{a,b,m}$ by forgetting these choices. It is closed in $Y_{a,b,m} \times Z$ as the incidence locus

$$H \subseteq \ker A_i, \qquad \operatorname{im} A_i \subseteq L_i \subseteq \ker B_i, \qquad \operatorname{im} B_i \subseteq L.$$

The projection π is consequently projective.

Every tuple in (8) lifts. If the stacked A is nonzero, take its common kernel hyperplane; otherwise choose any H. Similarly choose the common image line when B is nonzero, and any L otherwise.

For each branch, the rank bounds and $B_iA_i = 0$ allow a line between $\operatorname{im} A_i$ and $\ker B_i$. When all maps are nonzero these choices are unique and vary algebraically. Thus π is surjective and birational. Since $\widetilde{Y}$ is a vector bundle over a smooth irreducible variety, it is a resolution, and its image is irreducible. Each summand in (10) is a line, so

$$d = \dim Y_{a,b,m} = a + b + 3m - 2, \qquad D = \dim Z = a + b + m - 2. \tag{11}$$

The fiber over the zero tuple is exactly Z.

We show that π is small on $Y_{a,b,m} \setminus \{0\}$ when $a, b \le m$. Write $A = 0$ if all A_i vanish and similarly for B. Let k count indices with $A_i = B_i = 0$. When both A and B are nonzero, let s, t count indices with only A_i, respectively only B_i, zero. The fibers and codimensions of the resulting loci are

locus	fiber	codimension
$A \neq 0,\ B \neq 0$	$(\mathbb{P}^1)^k$	$s + t + 3k$
$A = 0,\ B \neq 0$	$\mathbb{P}^{a-1} \times (\mathbb{P}^1)^k$	$a + m + 2k - 1$
$A \neq 0,\ B = 0$	$\mathbb{P}^{b-1} \times (\mathbb{P}^1)^k$	$b + m + 2k - 1.$

For the first row the endpoint choices are fixed. In local trivializations, a branch has coordinates (u_i, φ_i) satisfying $\varphi_i(u_i) = 0$. The both-nonzero locus has dimension three, a locus with exactly one map zero has dimension two, and the zero tuple has dimension zero. This gives the first codimension formula. For the second row the nonzero rank-one matrix B has b rows and $2(m - k)$ potentially nonzero columns, so its locus has dimension $b + 2(m - k) - 1$. Subtracting this from d gives the stated codimension. The third row is symmetric.

Codimension minus twice fiber dimension is respectively

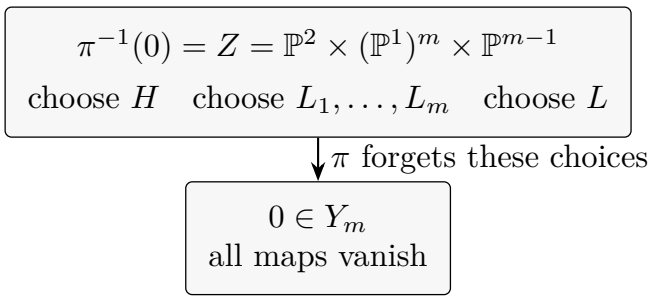
$$s + t + k, \qquad m + 1 - a, \qquad m + 1 - b.$$

It is strictly positive on every positive-dimensional-fiber locus away from zero when $a, b \le m$. These finitely many loci determine all fiber dimensions, proving the required smallness.

Decomposition and the ordinary polynomial

Now set $a = 3$, $b = m$, with $m \ge 3$. Then

$$d = 4m + 1, \qquad D = 2m + 1, \qquad Z = \mathbb{P}^2 \times (\mathbb{P}^1)^m \times \mathbb{P}^{m-1}.$$



$$b_2(Z) = 1 + m + 1 = m + 2,$$
$$\mu = b_2(Z) - 1 = m + 1.$$

Figure 5: At the origin all maps vanish, so every retained flag is free. The critical IC stalk is the next-to-top cohomology of this product after removing one class. Poincaré duality identifies its dimension with $b_2(Z) - 1 = m + 1$.

Let $K = R\pi_*\mathbb{Q}_{\widetilde{Y}}[d]$, and normalize IC_Y to restrict to $\mathbb{Q}[d]$ on the smooth open locus. Smallness over $Y \setminus \{0\}$ gives $K|_{Y\setminus\{0\}} \simeq \mathrm{IC}_{Y\setminus\{0\}}$. The decomposition theorem [12, Théorème 6.2.5] therefore gives IC_Y with multiplicity one, together with point-supported shifts only.

Proper base change identifies

$$\mathcal{H}^j(K)_0 = H^{j+d}(Z;\mathbb{Q}).$$

The largest nonzero stalk degree is $j = 1$, with dimension one. Since $\mathcal{H}^j(\mathrm{IC}_Y)_0 = 0$ for $j \geq 0$, exactly one $\mathbb{Q}_0[-1]$ occurs. Verdier self-duality of K and IC_Y forces exactly one $\mathbb{Q}_0[1]$. No further nonzero shifts can occur, by the upper stalk bound and self-duality. The zero shift is absent because d is odd and Z has no odd cohomology. Thus

$$R\pi_*\mathbb{Q}_{\widetilde{Y}}[4m+1] \simeq \mathrm{IC}_Y \oplus \mathbb{Q}_0[1] \oplus \mathbb{Q}_0[-1]. \tag{12}$$

We explicitly identify this calculation with the ordinary polynomial. For IC_{X_y} normalized to $\mathbb{Q}[\ell(y)]$ on its open Schubert cell, the Kazhdan–Lusztig theorem [13] gives

$$P_{x,y}(q) = \sum_{j\geq 0} \dim \mathcal{H}^{2j-\ell(y)}(\mathrm{IC}_{X_y})_x \, q^j.$$

The local product $U_x \times Y$ proved above shifts the stalk by $\ell(x)$, giving exactly

$$P_{x,y}(q) = \sum_{j\geq 0} \dim \mathcal{H}^{2j-d}(\mathrm{IC}_Y)_0 \, q^j. \tag{13}$$

There is no parabolic polynomial or change of parameter in this identity.

The even cohomology polynomial of Z is

$$\sum_{j\geq 0} \dim H^{2j}(Z;\mathbb{Q}) q^j = [3]_q[m]_q(1+q)^m.$$

The two point shifts in (12) lie in stalk degrees $-1, +1$. Under (13) these correspond to exponents $(d-1)/2 = 2m$ and $(d+1)/2 = 2m+1$. Subtracting them proves (3). Finally, the fiber polynomial is palindromic of degree $2m+1$, and its coefficient at q is $m+2$. Its coefficient at q^{2m} is therefore $m+2$, giving $\mu = m+1$ after subtraction. The length formulas in Theorem 1 follow from (9). This completes the proof.

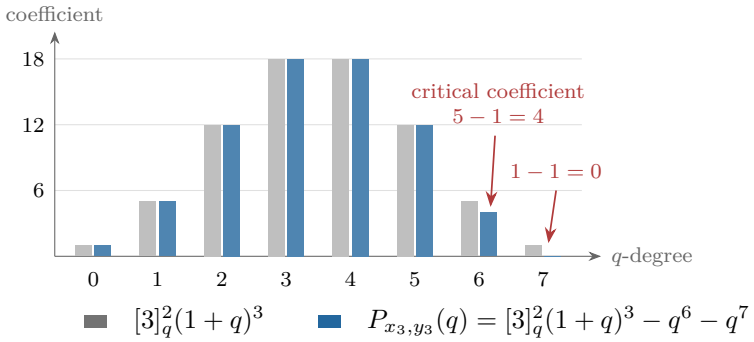


Figure 6: The first example. The raw fibre polynomial has coefficients $1, 5, 12, 18, 18, 12, 5, 1$. Removing the two point-supported shifts subtracts $q^6 + q^7$. The critical coefficient is therefore $5 - 1 = 4$.

The correction through McNamara's intersection form

There is a second way to determine the two point summands, using [9, Proposition 3.1], which attributes the multiplicity result to [14, Proposition 3.2]. For a smooth fiber Z of dimension D in a resolution of dimension d, the multiplicity of $\mathbb{Q}_0[t]$ is the rank of

$$H^{2D-d-t}(Z) \times H^{2D-d+t}(Z) \longrightarrow H^{2D}(Z), \qquad (\alpha, \beta) \longmapsto \alpha\beta e(E).$$

Here E is the normal bundle of the zero section and all coefficients are rational.

In our case $2D - d = 1$. Let w, z, u_i be the positive hyperplane classes on $\mathbb{P}^2$, $\mathbb{P}^{m-1}$, and the ith $\mathbb{P}^1$. Then

$$H^*(Z;\mathbb{Q}) = \frac{\mathbb{Q}[w, z, u_1, \dots, u_m]}{(w^3, z^m, u_1^2, \dots, u_m^2)}, \qquad \deg w = \deg z = \deg u_i = 2.$$

The two line summands for branch i in (10) have first Chern classes $-(w + u_i)$ and $-(z + u_i)$. Hence

$$e(E) = \prod_{i=1}^{m} (w + u_i)(z + u_i).$$

The integral of $w^2 z^{m-1} \prod_i u_i$ is one. To obtain all u_i factors in $we(E)$, one must choose $(w + z)u_i$ from each branch. Consequently

$$\int_Z we(E) = [wz^{m-1}](w + z)^m = m \neq 0.$$

Thus the pairing $H^0(Z) \times H^2(Z)$ and its transpose have rank exactly one. The $t = 0$ pairing vanishes because $H^1(Z) = 0$, and for $|t| \geq 2$ one factor is in negative degree. This gives exactly $\mathbb{Q}_0[1] \oplus \mathbb{Q}_0[-1]$, agreeing with (12).

3 Comparison with the other constructions

The three constructions answer different questions. The present pair is finite, its lower endpoint is the longest element of $S_3 \times S_{2m} \times S_m$, and its exact coefficient is $m + 1$. The exponential family of [21] has coefficient 2^r in S_{12+18r}, already in one left cell of shape $(4 + 6r)^3$. It gives faster growth; the present family gives every integer at least four and a closed formula for the entire ordinary polynomial. The two conclusions do not imply one another merely by comparing growth rates.

The affine family in the `Revision`, `Codex` and `GPT` folders is a different collection. It has period $2m + 1$ and the full finite-longest lower endpoint w_0. Its current web proof computes a row sum $\sum_j \mu_{j;m} = \binom{m+1}{2}$ rather than every individual polynomial. The fixed trivial source permits the first-cohomology application. No such application follows directly from the arbitrary finite pair of Theorem 1.

The contrast in the correction is especially simple:

Construction	Object that grows	Correction to control
Present finite family	The number of two-dimensional quiver branches; $b_2(Z) = m + 2$	Only $\mathbb{Q}_0[1] \oplus \mathbb{Q}_0[-1]$; smallness excludes additional supports away from zero
Finite exponential webs	The number of repeated web blocks	The correction is built into the repeated block before proving global canonicality
Affine row-sum webs	The thicknesses on one fixed annular graph	A full-spherical family must be removed; four other types miss the critical degree

The present resolution and the earlier affine geometric drafts share the idea of reading a critical stalk after subtracting other summands. Their varieties and permutations are different. Here the incidence conditions are rank-one maps with separate zero compositions; the affine drafts use square-zero endomorphisms with two flag-rank conditions.

There is also a concrete distinction from the three-row cell in [21]. The lower permutation x_m has longest increasing subsequence of length three: each of its three descending blocks contributes at most one entry, and three are attainable. The upper permutation contains the increasing

subsequence $2, 5, 6, N-1$. Their RSK shapes therefore differ, so the pair does not lie in one two-sided cell. For $m = 3$, direct insertion gives

$$\mathrm{shape}(x_3) = (3, 3, 3, 1, 1, 1), \qquad \mathrm{shape}(y_3) = (4, 4, 2, 2).$$

This explains why the existing rectangular invariant-web construction cannot simply be relabelled to give this pair.

4 Web transfer and the fixed-rank question

4.1 An exact finite Schur transfer

There is already a natural symmetric-web morphism space for the entire ordinary polynomial. It uses a source with $m+2$ packets, rather than a fixed-rank invariant tensor. Put

$$\gamma_m = (3, \underbrace{2, \ldots, 2}_{m \text{ times}}, m), \qquad \omega_N = (1, \ldots, 1), \qquad N = 3m + 3.$$

Let $\mathcal{H}_N$ be the finite type-A Hecke algebra with $(H_s + v)(H_s - v^{-1}) = 0$, and let C_w be its canonical basis in the normalization

$$C_w = \sum_{z \le w} v^{\ell(w)-\ell(z)} P_{z,w}(v^{-2}) H_z.$$

For W_γ permuting the consecutive source packets, write $C_\gamma = C_{w_\gamma}$ and $\mathcal{M}_\gamma = C_\gamma \mathcal{H}_N$. Finite Schur morphisms from γ_m to ω_N are $\mathrm{Hom}_{\mathcal{H}_N}(\mathcal{M}_{\gamma_m}, \mathcal{H}_N)$. In the symmetric-web presentation they are linear combinations of splits, merges and crossings, with no annular winding [7, 8].

For a maximal right-coset representative w, let A_w be its Schur index: $(A_w)_{i,k} = 1$ when the value i occurs in source packet k of w, and is zero otherwise. The normalized standard map $[A_w]_v$ sends C_γ to the standard sum on wW_γ whose coefficient at H_w is one. The canonical map is characterized by $\theta_{A_w}(C_\gamma) = C_w$. Explicitly,

$$[A_w]_v(C_\gamma) = \sum_{z \in wW_\gamma} v^{\ell(w)-\ell(z)} H_z.$$

Put $c_m(v) = v^{4m+1} P_{x_m, y_m}(v^{-2})$.

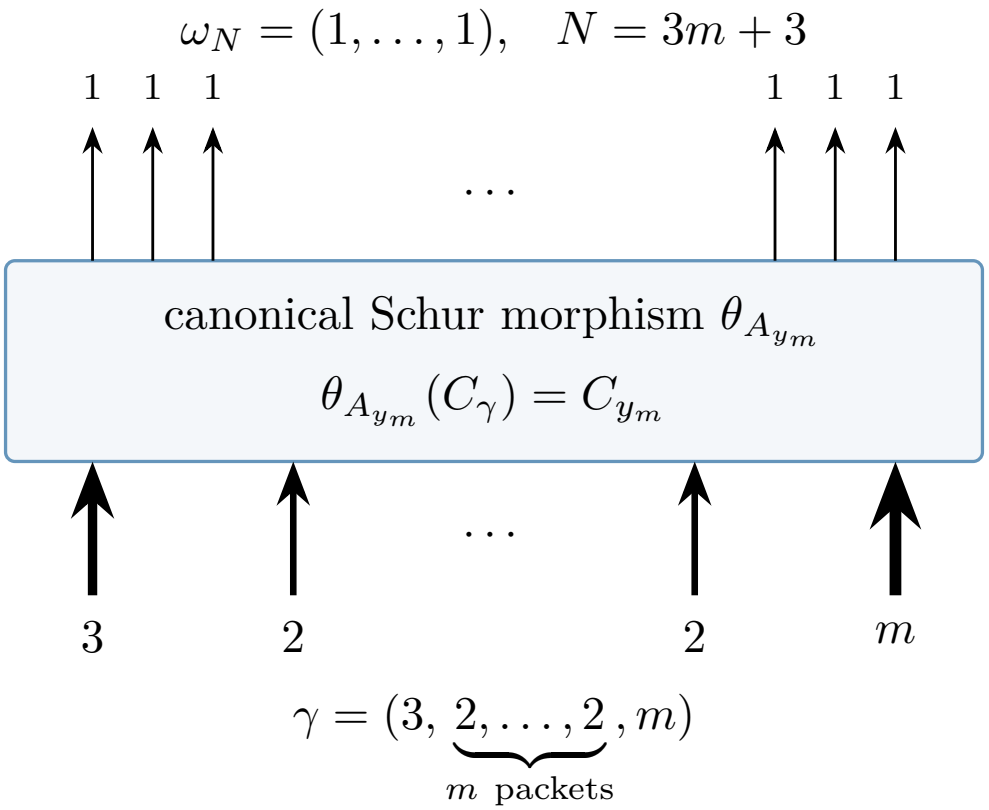
Proposition 3. *Both x_m and y_m are maximal right-coset representatives for W_{γ_m}. The corresponding canonical Schur coefficient is*

$$\boxed{[[A_{x_m}]_v]\theta_{A_{y_m}} = v^{4m+1} P_{x_m, y_m}(v^{-2}) = c_m(v).}$$

In particular its coefficient of v is $m+1$.

Proof. The source cuts are $3, 5, \ldots, 2m+3$. These are exactly the ascents of y_m, so every generator inside a source packet is a right descent. The only ascents of x_m are at 3 and $2m+3$, so it has all the same internal descents. This is precisely the maximal-coset condition.

The canonical descent identities put C_{y_m} in the required right-parabolic eigenspace, giving the canonical Schur map. Grouping its standard Hecke expansion by right cosets gives the normalized Schur standard expansion. Different cosets are disjoint, and the standard sum for A_{x_m} has coefficient one at H_{x_m}. Its coefficient is consequently the ordinary coefficient of H_{x_m} in C_{y_m}, as displayed. The length gap is $4m+1$; apply Theorem 1. □



$$\big[[A_{x_m}]_v\big]\theta_{A_{y_m}} = v^{4m+1} P_{x_m, y_m}(v^{-2})$$

$$[v]\big(\big[[A_{x_m}]_v\big]\theta_{A_{y_m}}\big) = m + 1.$$

Figure 7: The finite Schur transfer. The box denotes the canonical Schur morphism specified by its value on C_γ. It is a canonical linear combination of symmetric webs; the picture does not specify a new local web formula or an $\mathfrak{sl}_3$ invariant.

This proposition gives an exact home for the coefficient. It does not replace the geometric computation by a local web proof: writing C_{y_m} as a labelled box leaves its canonical correction to be computed. Also the total thickness N, the number $m+2$ of source packets, and a chosen quantum-group colour rank are separate parameters. Specializing a symmetric-web category to representations of $U_v(\mathfrak{sl}_3)$ requires proving that the relevant coordinate and its canonical normalization survive that specialization. In particular, the two point summands in the transverse slice do not by themselves give a global identity $\theta_{A_{y_m}} = W_m - (v+v^{-1})\theta_{A_{x_m}}$ for a specified raw Schur web W_m. Constructing that web and controlling all of its coordinates is a further mathematical step.

4.2 What would count as a solution?

The normalized polynomial to reproduce is

$$c_m(v) = v^{4m+1}P_{x_m,y_m}(v^{-2}) = v^{2m}\langle 3\rangle_v\langle m\rangle_v\langle 2\rangle_v^m - \langle 2\rangle_v, \qquad [v]c_m(v) = m+1, \tag{14}$$

where $\langle r\rangle_v = v^{r-1}+v^{r-3}+\cdots+v^{1-r}$. Beyond the abstract Schur transfer, there are two distinct goals. The stronger one is a specified coordinate of a single normalized dual-canonical tensor equal to $c_m(v)$. The weaker one is a new web construction whose coordinate has linear coefficient of v, with a proved ordinary-KL interpretation. Even the weaker goal requires global canonicality of the completed tensor. In the convention used below, the required conditions are

$$\overline{b_d} = b_d, \qquad b_d - e_d \in \bigoplus_a v\mathbb{Z}[v]e_a.$$

They concern every tensor coordinate, not only the selected one.

Here is the exact dictionary for the invariant SL_3 setting. For a word w with content (N,N,N), let $\pi(w)$ list positions of C, then B, then A, increasingly within each list. Let d be dominant, meaning every prefix has $\#A \geq \#B \geq \#C$, and let b_d be the normalized dual-canonical invariant. Put $x=\pi(a)$, $y=\pi(d)$ and $\delta = \ell(y)-\ell(x)$. The FKK identification [20, Theorems 1.3(2) and 2.5′], in the conventions of [21, Proposition 4A.1], says

$$[e_a]b_d = v^\delta \sum_{h\in W_J,\, xh\leq y} (-1)^{\ell(h)}P_{xh,y}(v^{-2}), \qquad W_J = S_N\times S_N\times S_N, \tag{15}$$

when $x\leq y$, and gives zero otherwise. The permutations are minimal right-coset representatives. To fix the parameter explicitly, set FKK's quantum variable $z=-v^{-1}$ and their polynomial parameter $Q=z^{-2}$. Their bar replaces Q by v^{-2}, and $(-z)^{-\delta}=v^\delta$; their second coproduct is the tensor-bar convention used here.

If $x=y$ the coefficient is 1. If $x<y$ and δ is even, parity excludes a term v. If δ is odd, every nonidentity summand has $xh\neq y$ and

$$\deg P_{xh,y} \leq \left\lfloor \frac{\delta-\ell(h)-1}{2}\right\rfloor \leq \frac{\delta-3}{2}.$$

Only $h=1$ can therefore contribute v, proving

$$\boxed{[v]\,[e_a]b_d = \mu(\pi(a),\pi(d)).} \tag{16}$$

Outside a strict Bruhat comparison the right side is interpreted as zero. If a is dominant as well, both permutations lie in the same left cell [21, Lemma 2C.5]. The full expression (15) is alternating parabolic; it must not be identified with the full ordinary polynomial without a further argument.

The present permutations are not directly of this three-letter form. Every $\pi(w)$ is a concatenation of three increasing subsequences, so it has no decreasing subsequence of length four. But x_m has a

descending block of length $2m$. Together with the different RSK shapes in Section 3, this rules out a direct reuse of the all-positive rectangular model. It is not an obstruction to every mixed-sign, clasped or otherwise transported $\mathfrak{sl}_3$ construction.

The usual longest-element symmetry does not immediately solve this problem either. It preserves μ under $(x, y) \mapsto (w_0 y, w_0 x)$ [15, Fact 6], but the displayed four-term increasing subsequence of y_m becomes a four-term decreasing subsequence of the new lower endpoint. It is still not a three-letter permutation. Lanini–McNamara can transport any interval into a single cell [10, Theorem 1.1], but their pattern embedding retains the original decreasing subsequences; it does not supply the required three-row rectangular shape. A different transport or a new canonical web construction is needed.

Scope of the reported computational obstructions. The source note refers to `web_realization_obstructions.zip`, but that archive was not among the supplied files. The local-matrix, pole-rank and exhaustive readout assertions below have therefore not been independently reproduced in this assessment. Their consequences are conditional on those local identities. None is used in the geometric proof or Proposition 3.

4.3 Strategy A: transport the geometric canonical correction

The decomposition (12) already isolates the desired canonical sheaf. There is an exact quiver realization of this statement. Take the quiver with vertices $A, B, 1, \dots, m$ and arrows $A \to i \to B$, of dimension vector $(3, 2, \dots, 2, m)$. In the increasing stable-flag convention, put

$$\Xi = F_1 \cdots F_m, \qquad M_m = F_A^{(2)} \Xi F_A F_B \Xi F_B^{(m-1)}, \qquad D_m = F_A^{(3)} \Big(\prod_i F_i^{(2)} \Big) F_B^{(m)}.$$

The middle generators commute. The stable flags defining M_m impose exactly the incidence conditions of (10); the flag for D_m forces all arrows to be zero. Normalized convolution and Lusztig's construction [17, 18] therefore turn (12) into the canonical-basis relation

$$b_m = M_m - (v + v^{-1}) D_m.$$

Here b_m corresponds to IC_{Y_m} and D_m to the zero-orbit sheaf. With shifts represented by v, the additive stalk character $\epsilon_0(\mathcal{F}) = \sum_j \dim \mathcal{H}^j(\mathcal{F})_0 v^{-j}$ satisfies $\epsilon_0(b_m) = c_m(v)$. No multiplicativity of ϵ_0 is asserted.

The correction also has an explicit geometric projector. For the classes used in Section 2,

$$\int_Z we(E) = m, \qquad \int_Z u_1 e(E) = m - 1, \qquad \alpha = w - u_1, \quad \int_Z \alpha e(E) = 1.$$

Writing $K = R\pi_* \mathbb{Q}[d]$ and $\delta_0 = \mathbb{Q}_0$, let $g : \delta_0 \to K$ be the degree -1 map from the class $1 \in H^0(Z)$ and let f be its dual, also of degree -1. Nonequivariantly the self-intersection formula gives $fg = 0$, $f\alpha g = 1$ and $f\alpha^2 g = 0$. Hence

$$P = gf\alpha + \alpha gf, \qquad P^2 = P, \qquad \mathrm{im}(1 - P) = \mathrm{IC}_{Y_m}.$$

Indeed $(g, \alpha g)$ and $(f\alpha, f)^T$ split $\delta_0[1] \oplus \delta_0[-1]$ from K.

Remaining step. Realize this correction in a fixed-rank $\mathfrak{sl}_3$ model by a functor or explicit web maps preserving the relevant canonical normalization. The quiver's underlying graph is $K_{2,m}$, not a type-A Dynkin diagram; the canonical-basis statement is not automatically an SL_3 tensor statement. Also $2 \dim Z = d + 1$, so this resolution is not semismall at zero and cannot be substituted directly into a semismall Satake argument.

4.4 Strategy B: upgrade a clasped multiplicity model

There is a concrete fixed-rank model with the correct dimension. For the defining quantum SL_3 module V, set

$$T_m = \mathrm{Sym}_v^2 V \otimes \mathrm{Sym}_v^{m-1} V \otimes V^{\otimes m}, \qquad L_m = S_v^{(2m,1)} V.$$

Then

$$\dim \mathrm{Hom}_{U_v(\mathfrak{sl}_3)}(L_m, T_m) = m + 1. \tag{17}$$

One proof uses the $m+2$ tensor blocks of sizes $(2, m-1, 1, \ldots, 1)$. The weight $(2m, 1, 0)$ has one basis vector for lowering each block once; the highest-weight condition imposes one nonzero linear equation. This gives $m+1$ explicit highest vectors. The spider equivalence and its idempotent completion [19] express the corresponding intertwiners as clasped-web combinations.

There is also a geometric vector-space match. The critical stalk is

$$\mathcal{H}^{-1}(\mathrm{IC}_{Y_m})_0 \simeq H^{4m}(Z; \mathbb{Q})/\mathbb{Q}e(E).$$

Choose the ample class $\mathcal{L} = 2(m-1)(w+z) + m\sum_i u_i$. The Euler pairings show $e(E)$ is a nonzero multiple of $\mathcal{L}^{2m}$. Hard Lefschetz identifies the displayed quotient with the primitive part of $H^2(Z)$, with basis

$$(m-1)w - 2z, \quad 2z - mu_1, \quad m(u_1 - u_2), \ldots, m(u_{m-1} - u_m).$$

These can be matched with the highest vectors at $v = 1$ after fixing normalizations.

Remaining step. The $m+1$ independent intertwiners form a multiplicity space. They are not one tensor coordinate of one canonical vector. One must construct an appropriate canonical tensor and prove that its chosen coordinate counts these contributions. Clasping or matching dimensions alone does not establish this; a canonical-basis-compatible inclusion or quotient must be justified separately.

4.5 Strategy C: complete an additive web transfer canonically

The most concrete current lead uses the corrected 28-state block of Goertzen–Lacabanne–Tubbenhauer–Vaz [21]. Its interface is $V^{\otimes 3} \oplus \mathbf{1}$. Number triple states lexicographically from 0 to 26, and use 27 for the scalar summand. For selected external word $J_* = CBABACACB$, let $A(v)$ be the gauged block matrix. Exact expansion gives the complete first two matrices

$$A_0 = [v^0]A = E_{11} + E_{33}, \qquad A_1 = [v]A = 2E_{13} + 2E_{31} + E_{39} + E_{93}, \tag{18}$$

where $1 = AAB$, $3 = ABA$ and $9 = BAA$. Every unlisted entry in A_0, A_1 is zero. Thus

$$\boxed{[v](A(v)^k)_{1,3} = [v](A(v)^k)_{3,1} = 2k \quad (k \geq 0).} \tag{19}$$

For $k \geq 1$, use

$$[v]A(v)^k = \sum_{j=0}^{k-1} A_0^j A_1 A_0^{k-1-j}.$$

Each summand has the indicated entries equal to 2. This proves linear growth for every k; it is not extrapolation from computed powers.

Why this is not yet a KL family. The displayed boundary states are literal coordinate vectors, not established canonical endpoints. The existing prefix already satisfies

$$p(0)e_{AAB} = e_{ABAABC} + e_{ABACAB}, \qquad p(0)e_{ABA} = e_{ABABAC} + e_{ABCAAB}.$$

After one block each has 24 distinct constant output coordinates. A normalized dual-canonical tensor has exactly one constant standard coordinate. Because the gauged maps have no negative powers of v, positive-degree corrections to these input vectors cannot remove the extra constants.

Remaining step. Construct endpoint or canonical correction maps that remove these unwanted constant terms while retaining the contribution in (19). Only after proving bar invariance, integrality and the unique-leading-term condition for the completed tensor can (16) be applied. At present there is no new web-based KL family of slope 1 or 2 established by this calculation.

4.6 Obstructions that narrow these strategies

A rational realization of the full target fails integrality. An explicit bar-fixed rational map $F : V \otimes V^* \to (V \otimes V^*) \otimes V^{\otimes 3}$, with a six-leg rational preparation Q and the standard trace cap C, gives the coordinate $c_m(v)$ on $(123)^{m+2}$. However, other tensor coordinates have poles of order $2m-2$ at $v = i$. More strongly, put $B = 1 + v^2$ and let s be the minimum B-valuation of any fixed nonzero rational preparation Q', allowing any fixed auxiliary boundary. For this particular F, C,

$$\mathrm{val}_B(CF^kQ') = s + 1 - 2k \qquad (k \geq 2).$$

The uniform proof uses exact full-tensor ranks $9, 8, 9$ for $R(i)$, $(CR/B)(i)$ and $(((CR/B)(i)) \otimes 1)R(i)$, respectively, where $R = (v+v^{-1})^2F$. The first and third maps are injective. Thus changing only a fixed preparation cannot repair this construction. A separate calculation excludes integral triangular replacement matrices with eigenvalues a permutation of $v^s(1, 1+v^2, v^2(1+v^2))$ in the specified channels, for any integer s. Neither statement is a universal obstruction to web realizations.

Changing the readout of the fixed GLTV family is insufficient. GLTV's canonical label is

$$d_k = SM^kT, \quad S = ABAAAC, \quad M = BBBAACACC, \quad T = BBCBCC.$$

Our exact local certificates and a uniform argument show that

$$[v]\,[e_a]b_{d_k} \in \{0, 1, 2, 4, 8, \ldots\} \quad \text{for every selected word } a \text{ and every } k. \tag{20}$$

The mechanism is restrictive: each constant local matrix has at most one nonzero entry per row, with weights 1 or 2. Apart from the canonical tail, a first-order tail selects one coordinate. For the canonical tail, only the rightmost selected block different from M can supply degree one. Each surviving term therefore follows one path and multiplies weights; there is no sum over insertion positions. In particular, our value $\mu = 5$ cannot occur by merely selecting a different coordinate of any b_{d_k}.

The source describes these as computer-assisted proofs from exhaustive finite local identities and arguments valid for every iteration count, and reports two independent implementations of the constant and linear GLTV coefficients. Those certificates were not supplied for this assessment. The new statements are not claimed to have been formally verified in Lean. They leave open new canonical labels, other endpoint constructions, modified blocks, and transport from the geometric model.

4.7 Precise current status

The ordinary finite-type family and its full polynomial are proved in Section 2, giving every integer at least four; known small examples complete Corollary 2. Proposition 3 gives an exact canonical coefficient in the finite symmetric-web category, without a new local computation of the canonical morphism. The quiver correction and clasped multiplicity model provide further bridges, without yet identifying a three-colour invariant-web coordinate. The four-colour route in Section 5 instead transports the coefficient through bent Hecke operators; its canonical and leading-label identifications remain explicit inputs. Conditional on the stated local matrices,

formula (19) supplies an additive mechanism; its canonical completion remains unresolved. Thus the next concrete problem is to preserve that additive contribution while constructing and proving the appropriate global canonical tensor.

5 Four-row canonical labels and hourglass graphs

The report [1] proposes a different transfer of the same linear coefficient. Instead of realizing the entire polynomial in a three-colour web, it transports one Hecke multiplication coefficient to a four-colour invariant tensor. Bending an operator turns its input boundary into dual tensor factors; replacing each dual factor by an exterior cube then gives an all-positive boundary. The resulting rectangular labels are explicit. Their canonical interpretation uses the bending and leading-label lemmas of the local package cited in the report, so we state that part conditionally.

5.1 The Hecke coefficient to transport

Keep $N = 3m + 3$ and complement and interchange the original endpoints:

$$X_m = w_0 y_m, \qquad Y_m = w_0 x_m.$$

Longest-element symmetry preserves the ordinary μ-coefficient [15, Fact 6], hence $\mu(X_m, Y_m) = m + 1$. This uses a symmetry of μ, not an equality of the two full KL polynomials. The reported RSK shapes are

$$\mathrm{sh}(X_m) = (m+1, m+1, m-1, 2), \qquad \mathrm{sh}(Y_m) = (2m, m, 3).$$

They explain the choice of four colours. Both endpoints survive the rank-four Schur–Weyl quotient, whereas X_m is killed at rank three. They still have different shapes, so another step is needed to put them in a common rectangular cell.

The required row bounds have a direct uniform proof. The entries of y_m partition into four decreasing subsequences:

$$\begin{gathered}(h, h-2, h-4, \ldots, 5, 1), \qquad (3, 2),\\ (N, h-1, h-3, \ldots, 6, 4), \qquad (N-1, N-2, \ldots, h+1).\end{gathered}$$

Its increasing subsequence $(2, 5, 6, N-1)$ has length four. Hence its longest increasing subsequence has length exactly four. Similarly, x_m is the concatenation of three decreasing blocks, with values increasing from one block to the next, so its longest increasing subsequence has length three. Complementation turns these into the longest decreasing subsequences of X_m, Y_m. Schensted's theorem therefore gives exactly four and three rows, respectively, without needing the full shape formulas.

Let E be the natural $U_v(\mathfrak{sl}_4)$-module. In the Hecke normalization of Section 4.1, put $K_s = v^{-1} - H_s$, and obtain K_w from C_w by the automorphism $H_s \mapsto -H_s^{-1}$. On the ordered pair of words (ab, ba) with $a < b$, the tensor operator is

$$K_s = \begin{pmatrix} v^{-1} & -1 \\ -1 & v \end{pmatrix}, \qquad K_s(aa) = 0.$$

The explicit endpoints satisfy $X_m s_5 < X_m$ and $Y_m s_5 > Y_m$. The KL multiplication rule therefore puts

$$(m+1)K_{X_m} \quad \text{in the expansion of } K_{Y_m} K_{s_5}. \tag{21}$$

This is the coefficient that the transfer retains. It need not retain the full polynomial in Theorem 1.

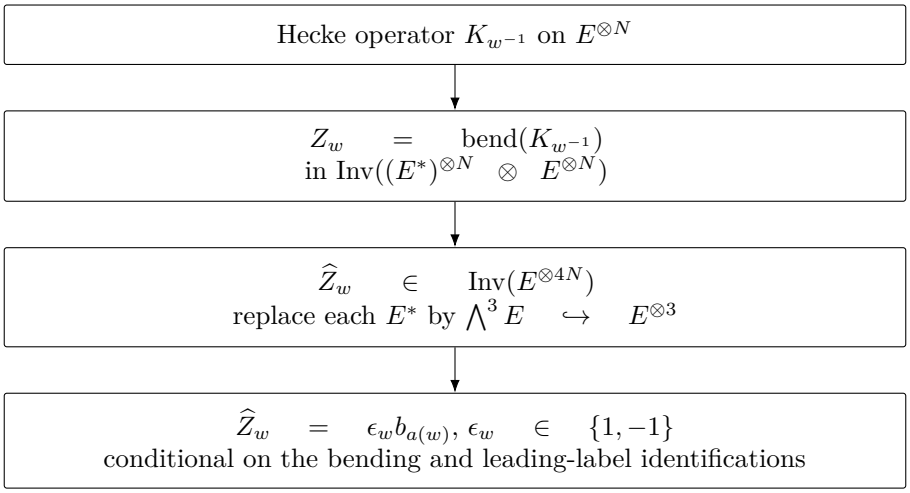


Figure 8: The proposed rank-four transfer. Bending changes an operator into an invariant tensor. The last identification is the substantive canonical-basis input; it is not inferred from a planar drawing.

5.2 An explicit algorithm for the rectangular labels

Use colours $0, 1, 2, 3$, printed as A, B, C, D. For a standard tableau T of shape λ, let T_{row} be the tableau filled consecutively along its rows, and set

$$c_\lambda = 0^{\lambda_1} 1^{\lambda_2} 2^{\lambda_3} 3^{\lambda_4}, \qquad z(T) = \mathrm{RSK}^{-1}(T_{\mathrm{row}}, T), \qquad F(T)_j = (c_\lambda)_{z(T)(j)}.$$

Empty rows contribute no letters. For $(P, Q) = \mathrm{RSK}(w)$, first form the mixed word

$$L(w) = \mathrm{rev}\, F(P) \mid \mathrm{comp}\, F(Q), \qquad \mathrm{comp}(i) = 3 - i.$$

The first N letters label dual factors. Replace each of them by the increasing triple omitting its complementary colour:

dual letter	A	B	C	D
replacement	ABC	ABD	ACD	BCD.

Keep the N letters after the divider. The resulting word $a(w)$ has length $4N$, with the divider now after position $3N$. Finally, $\pi(a)$ lists positions of D, then C, then B, then A, increasingly within each group. Define

$$U_m = \pi(a(X_m)), \qquad V_m = \pi(a(Y_m)). \tag{22}$$

This algorithm defines the endpoints for every $m \geq 3$, without a search for a favourable coefficient. The claim that it gives the leading coordinates of the bent tensors is a separate lemma. The report relates it to Hatt's labelled cell modules [5, equation (107), Theorem 167, Corollary 169]; the translation of the duality and RSK conventions is part of the required input.

For example, when $m = 3$, the two words are

$$a(X_3) = \texttt{ABC ABC ABC ABD ABD ACD ABD ACD ABC BCD BCD ABD} \mid \texttt{CAADBCBCCDDD},$$
$$a(Y_3) = \texttt{ABC ABC ABC ABD ABD ABD ABC ABC ABC ACD ACD ACD} \mid \texttt{BBBDDDCCCDDD}.$$

Each contains twelve copies of each colour. Every prefix satisfies $\#A \geq \#B \geq \#C \geq \#D$. Such a dominant balanced word labels a four-row rectangular tableau: put the positions of each colour in one row. For $\pi(a)$ this is the insertion tableau, while the recording tableau is row-superstandard. Thus dominant balanced labels give a common left cell.

5.3 What is needed to identify a canonical tensor

The report's bending argument uses the based right dual and cup

$$b_i^* = (-v^{-1})^{4-i} e_i^*, \qquad \mathrm{cup}(1) = \sum_{i=1}^{4} (-v)^{4-i} b_i^* \otimes e_i.$$

For the integral Laurent mixed coordinates $f_a(v)$ of Z_w, the norm calculation required by that argument is

$$\sum_a f_a(v)^2 = \sum_{\substack{\lambda \vdash N \\ \ell(\lambda) \le 4}} s_\lambda(1, v^2, v^4, v^6)\, \chi_\lambda(K_{w^{-1}} K_w). \tag{23}$$

Here s_λ is a Schur polynomial and χ_λ the corresponding Hecke character. This is a weighted-trace identity in specified coordinates, rather than a claim that ordinary dot product is an invariant form.

The proposed proof combines the initial power $v^{2n(\lambda)}$, where $n(\lambda) = \sum_i (i-1)\lambda_i$, in the Schur polynomial with Geck's balanced representations and leading matrix units [4, Section 3.2, Lemma 3.4, Corollary 5.6]. The convention conversion is $T_s^{\mathrm{Geck}} = H_s^{-1}$ and $C_w^{\mathrm{Geck}} = (-1)^{\ell(w)} K_w$, with transposed partition indices. The claimed result for a surviving operator is a norm $1 + O(v)$.

This has a useful elementary consequence. A negative least valuation among the f_a would give a nonzero sum of integer squares in a negative degree of the norm. Hence all coordinates are regular at zero, and exactly one has constant coefficient ± 1. With bar invariance and polynomial unitriangularity this identifies a dual-canonical tensor up to sign. The exterior-power inclusion is compatible with the relevant canonical bases [3, Section 5, Theorem 11]. Locating the unique constant coordinate still requires the leading-label lemma; the norm does not supply its word.

The distinction between a Gram-dual basis change and an abstract dual representation matters here. A basis change congruent to the identity does not itself introduce a new duality; the inverse in $K_{w^{-1}}$ accounts for the reversed input boundary. The local orientation audit cited in the report is needed to check this normalization against its label algorithm.

Proposition 4 (Conditional rank-four transfer)**.** *Assume the bent-canonical and leading-label identifications described in [1, 2], with the duality above: the surviving K_w give distinct signed dual-canonical tensors $\widehat{Z}_w = \epsilon_w b_{a(w)}$, and the multiplication in (21) is transported to the indicated local Hecke operator. Then, for every $m \ge 3$,*

$$U_m, V_m \in S_{12m+12}, \qquad Q(U_m) = Q(V_m), \qquad U_m < V_m, \qquad \mu(U_m, V_m) = m + 1.$$

Proof. The multiplication coefficient in (21) is $m + 1$. Under the assumed identifications, bending and exterior inclusion retain it, up to the signs of the two normalized canonical vectors. The local operator acts at positions $3N + 5, 3N + 6$ of the all-positive boundary.

To compare with ordinary KL theory, put $u = -v$ and $h_s = H_s^{-1}$. Then $C'_s = h_s + u = -K_s$. The weight space of content $(N)^4$ is the antispherical module for $J = S_N^4$, with minimal representatives $\pi(a)$. The signed FKK dictionary [20, Theorems 1.3(2), 2.5′] identifies multiplication coefficients between distinct retained canonical vectors with ordinary μ-coefficients, apart from a possible top term of coefficient one. Our coefficient has absolute value $m + 1 > 1$, so it is not the top term. Positivity fixes its sign, and the multiplication rule gives the strict Bruhat comparison. The dominant balanced labels give the common row-superstandard recording tableau. □

One can also see why the coefficient is ordinary directly. For $\delta = \ell(\pi(b)) - \ell(\pi(a))$, the signed tensor coordinate is

$$[e_a] b_b = u^\delta \sum_{\substack{t \in J \\ \pi(a) t \le \pi(b)}} (-1)^{\ell(t)} P_{\pi(a)t, \pi(b)}(u^{-2}). \tag{24}$$

When δ is odd and the minimal representatives are distinct, every $t \neq 1$ has ordinary KL degree below $(\delta-1)/2$. Consequently $[u][e_a]b_b = \mu(\pi(a), \pi(b))$. The full coordinate is parabolic; this particular coefficient is ordinary. Here $u = -v$, so extracting $[v]$ in the raw signed convention reverses the sign. This parameter change must be retained when comparing with the positive-colour convention of (16).

5.4 Repeated words and exact finite checks

The report gives compact block expressions for the algorithmic labels. Write $r = m-3$ and define the following words independently:

$$\begin{aligned}\alpha_m &= (\mathtt{ABC})^3(\mathtt{ABD})^2\mathtt{ACDABDACDABC}(\mathtt{ABDABC})^r \\ &\qquad (\mathtt{BCD})^2(\mathtt{ACD})^r\mathtt{ABD} \mid \mathtt{CAAD}(\mathtt{BC})^{m-1}\mathtt{CD}^m, \\ \beta_m &= (\mathtt{ABC})^3(\mathtt{ABD})^3(\mathtt{ABC})^{2m-3}(\mathtt{ACD})^3(\mathtt{ABD})^r \\ &\qquad \mid \mathtt{BBBD}^m\mathtt{C}^m\mathtt{D}^m.\end{aligned}$$

The divider is not a letter. Independent RSK checks give $\alpha_m = a(X_m)$ and $\beta_m = a(Y_m)$ for $m = 3, \ldots, 30, 50, 100$. An all-parameter derivation of these block identities by row insertion remains to be written. The conditional construction in Proposition 4 uses the algorithmic labels, so it does not assume this simplification.

There is an exact symbolic calculation for the displayed block words. If $c_i(B)$ counts colour i in B and $A(B)$ counts pairs $p < q$ with $B_p < B_q$, then

$$\begin{aligned}\ell(\pi(B)) &= A(B), \\ A(BC) &= A(B) + A(C) + \sum_{i<j} c_i(B)c_j(C), \\ A(B^k) &= kA(B) + \binom{k}{2}\sum_{i<j} c_i(B)c_j(B).\end{aligned}$$

Substitution proves, without interpolation, that both words have N copies of each colour and

$$\ell(\pi(\alpha_m)) = \frac{69m^2+125m+112}{2}, \qquad \ell(\pi(\beta_m)) = \frac{77m^2+141m+90}{2}.$$

Their length difference is $4m^2+8m-11$. At the checked parameters these are the lengths of U_m, V_m:

m	$4N$	$\ell(U_m)$	$\ell(V_m)$	$(\ell(V_m)-\ell(U_m)-1)/2$	local positions
3	48	554	603	24	41, 42
4	60	858	943	42	50, 51
5	72	1231	1360	64	59, 60

The local colours are BC in $a(X_m)$ and DD in $a(Y_m)$. The operator acts on the canonical tensor, not just on its leading word, so the equal colours in the latter do not make the whole calculation vanish.

The independent checker `assessment/check_report.py` verifies RSK and inverse RSK, balanced dominance, the common recording tableau, strict Bruhat order, lengths and these local colours at all thirty listed values. It also checks the block statistics symbolically over $\mathbb{Q}[m]$. These computations do not evaluate a rectangular KL polynomial or prove the bending lemmas.

5.5 What the hourglass diagrams show

The three plates following the references reproduce the six growth representatives in [1], for $m = 3, 4, 5$. They use the four-row hourglass construction of Gaetz–Pechenik–Pfannerer–Striker–Swanson [6, Algorithm 5.1, Theorems 5.4 and 6.1]; the exterior-power spider framework is supplied

by [19]. Each internal vertex has weighted degree four. A teal edge marked 2 is a double hourglass, whose multiplicity encodes its prescribed twist. Red boundary positions mark the local Hecke operator.

These are growth representatives for the corresponding tableau classes. They are not asserted to be distinguished top representatives or to evaluate to the transferred dual-canonical tensors. In particular, benzene moves can preserve the leading label while changing the tensor invariant [6, Remark 6.5]. The coefficient in Proposition 4 comes from the canonical transfer, not from evaluating these pictures.

The report records further tests: a 92-file manifest and ten reproduction commands, all 576 ordered pairs in S_4, 192 signed bending tests, ordinary KL recursions for the original members $m = 3, 4, 5$, and graph trip permutations compared with promotion. The accompanying package, orientation audit and `audit.json` were not supplied with `report.tex`. Those results remain reported evidence. The available label and embedded graph data are checked independently in `assessment/`; these checks do not replace the missing tensor or KL computations.

There is also an earlier searched route, with reported common-cell shapes $(6, 5, 4, 3)$, $(8, 7, 7, 4)$ and $(10, 9, 8, 5)$ for $m = 3, 4, 5$, completed to $(6)^4$, $(8)^4$ and $(10)^4$. The report describes finite Warrington move certificates and identical-tail deletions for that route. It gives no uniform move sequence connecting these pairs with the new rectangles $(12)^4$, $(15)^4$ and $(18)^4$. Equality of coefficients alone does not identify the two constructions.

The next tasks are specific: derive the repeated block labels uniformly; check the bending and orientation lemmas against the cited based-module results; and compare a specified hourglass representative with the transferred canonical tensor. A repeated boundary word does not yet give a fixed-width graph interface or a local additive graph rule. The four-row construction also leaves the original three-colour problem open.

A One initial example and a permutation generator

For $m = 3$ the pair in S_{12} is

$$x_3 = (3, 2, 1, 9, 8, 7, 6, 5, 4, 12, 11, 10), \qquad y_3 = (9, 3, 2, 12, 7, 8, 5, 6, 1, 11, 10, 4).$$

Its lengths are $21, 34$, and

$$P_{x_3,y_3}(q) = 1 + 5q + 12q^2 + 18q^3 + 18q^4 + 12q^5 + 4q^6.$$

The original note reports an independent exact ordinary-KL recurrence check of this polynomial. The later report records the same check for $m = 4, 5$, giving

$$\begin{aligned} P_4 &= 1 + 6q + 17q^2 + 31q^3 + 41q^4 + 41q^5 + 31q^6 + 17q^7 + 5q^8, \\ P_5 &= 1 + 7q + 23q^2 + 48q^3 + 73q^4 + 88q^5 + 88q^6 \\ &\qquad + 73q^7 + 48q^8 + 23q^9 + 6q^{10}. \end{aligned}$$

They agree with Theorem 1. The recurrence implementation was not supplied with the report. The reproducible checks in `assessment/` verify permutations, lengths, essential corners, Bruhat comparisons, source descents, RSK insertion and the arithmetic of the proposed polynomial. They do not implement a second KL recurrence. The all-m conclusion rests on the geometric proof above.

The following standard Python code generates the one-based permutations.

```
def family(m):
    assert isinstance(m, int) and m >= 3
```

```
    h, N = 2*m + 3, 3*m + 3
    x = [3, 2, 1] + list(range(h, 3, -1))
    x += list(range(N, h, -1))
    pairs = [v for j in range(m, 1, -1)
               for v in (2*j + 1, 2*j + 2)]
    y = [h, 3, 2, N] + pairs + [1]
    y += list(range(N - 1, h, -1)) + [4]
    assert sorted(x) == sorted(y) == list(range(1, N + 1))
    return tuple(x), tuple(y)
```

The source note refers to a separate computational archive, `web_realization_obstructions.zip`, containing the local certificates, two implementations, and their source hashes; it was not present in the supplied folder. The GLTV port uses upstream commit

```
5dbb3ed519677c4c81ec8a4f8fd46cd56d355a40
```

of https://github.com/dtubbenhauer/mu. The present geometric proof does not depend on that archive.

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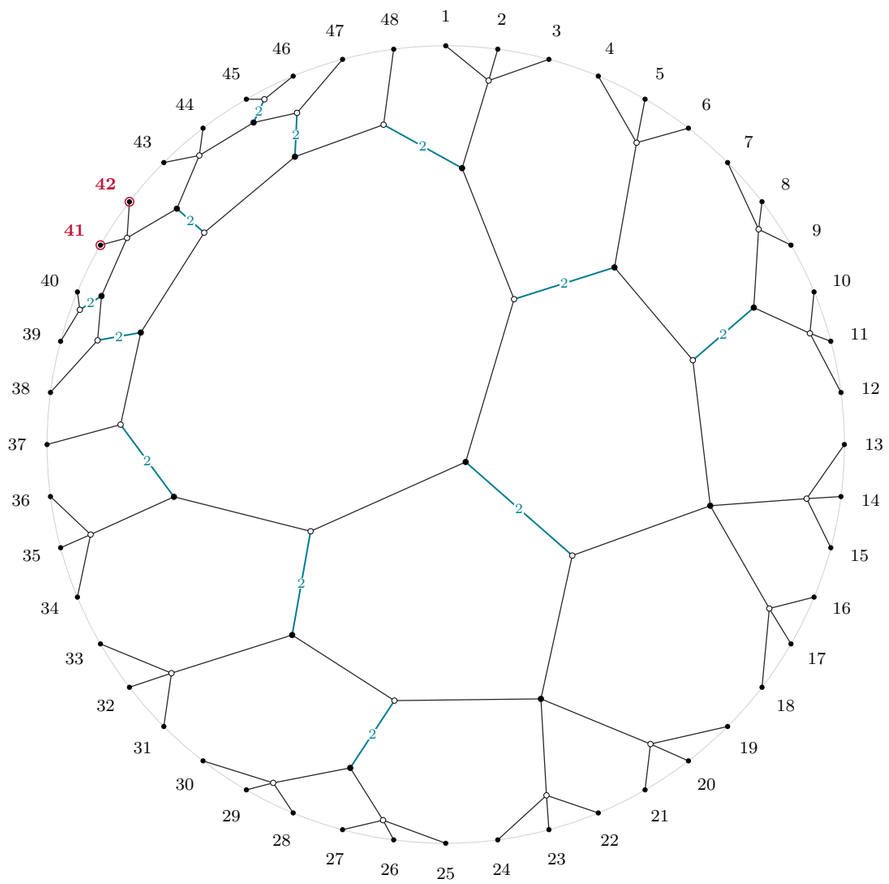
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Uniform four-row labels: $m = 3$, shape $(12)^4$

Transfer formula: $\mu = 4$ S_{48} Length difference 49 Critical q-degree 24

Source: $\ell = 554$

40 internal vertices; 12 double edges

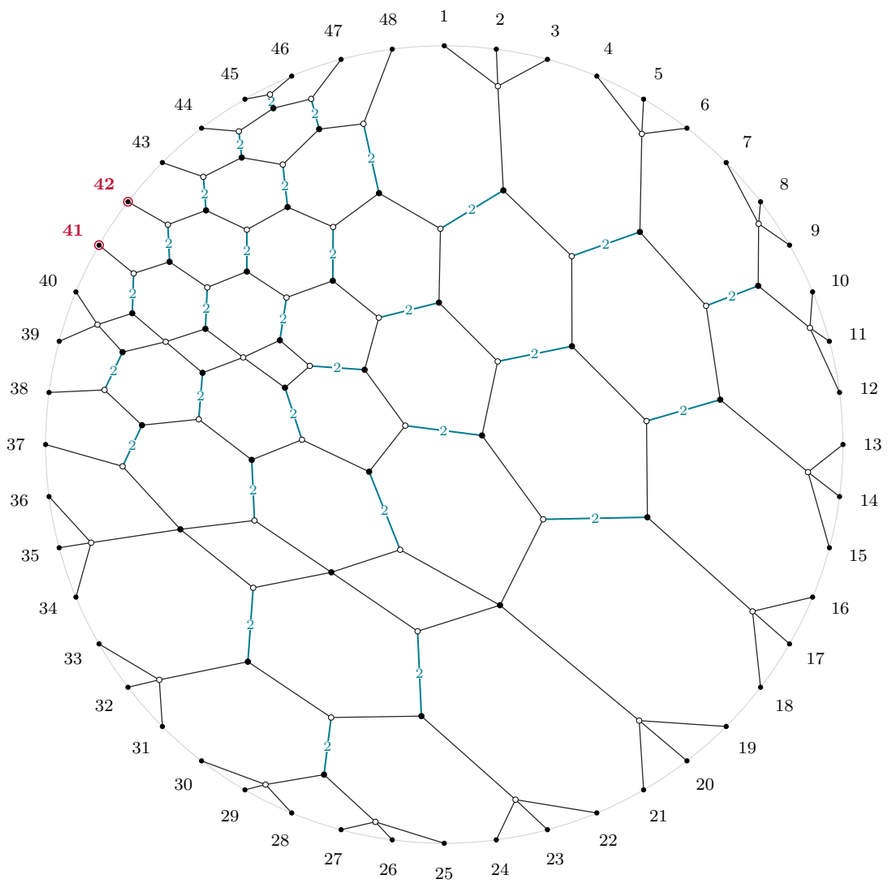


Tableau

1	4	7	10	13	16	19	22	25	34	38	39
2	5	8	11	14	20	26	28	31	35	41	43
3	6	9	17	23	27	29	32	37	42	44	45
12	15	18	21	24	30	33	36	40	46	47	48

Target: $\ell = 603$

78 internal vertices; 30 double edges



Tableau

1	4	7	10	13	16	19	22	25	28	31	34
2	5	8	11	14	17	20	23	26	37	38	39
3	6	9	21	24	27	29	32	35	43	44	45
12	15	18	30	33	36	40	41	42	46	47	48

Red positions: $41, 42$; source colors BC, target colors DD.

Teal edges labelled 2 denote double hourglasses. The boundary circle is a drawing guide.

Growth representatives for the canonical labels; not asserted equal to the canonical tensors.

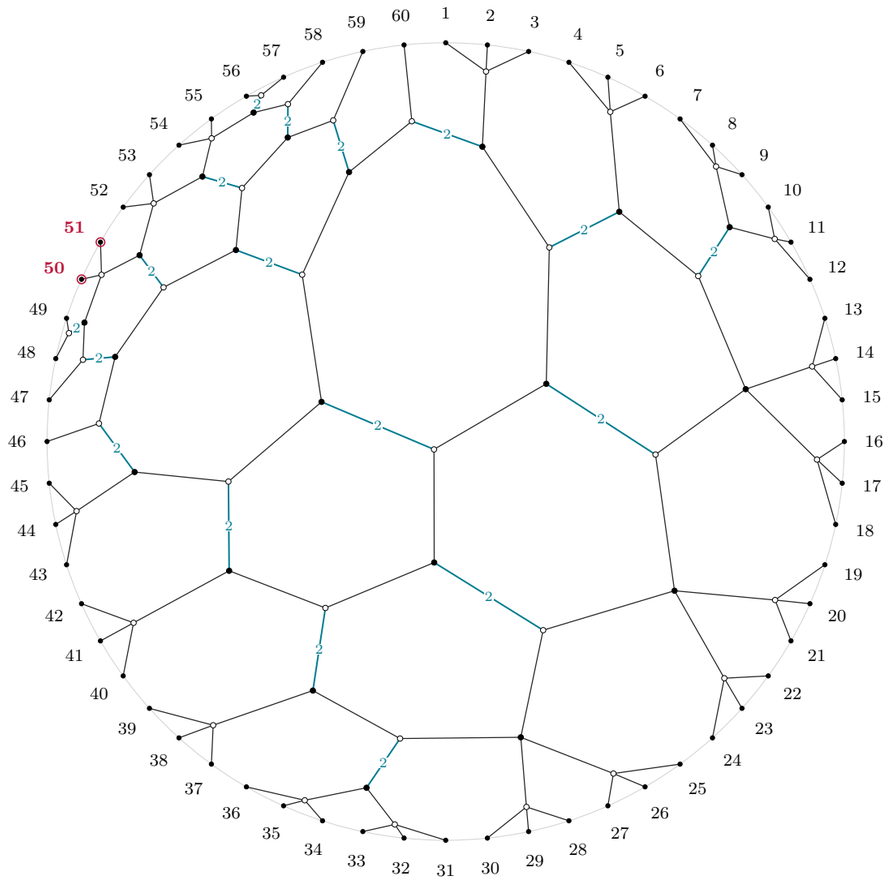
The coefficient is supplied by the transfer argument, not evaluated from these diagrams.

Uniform four-row labels: $m = 4$, shape $(15)^4$

Transfer formula: $\mu = 5$ S_{60} Length difference 85 Critical q-degree 42

Source: $\ell = 858$

57 internal vertices; 18 double edges

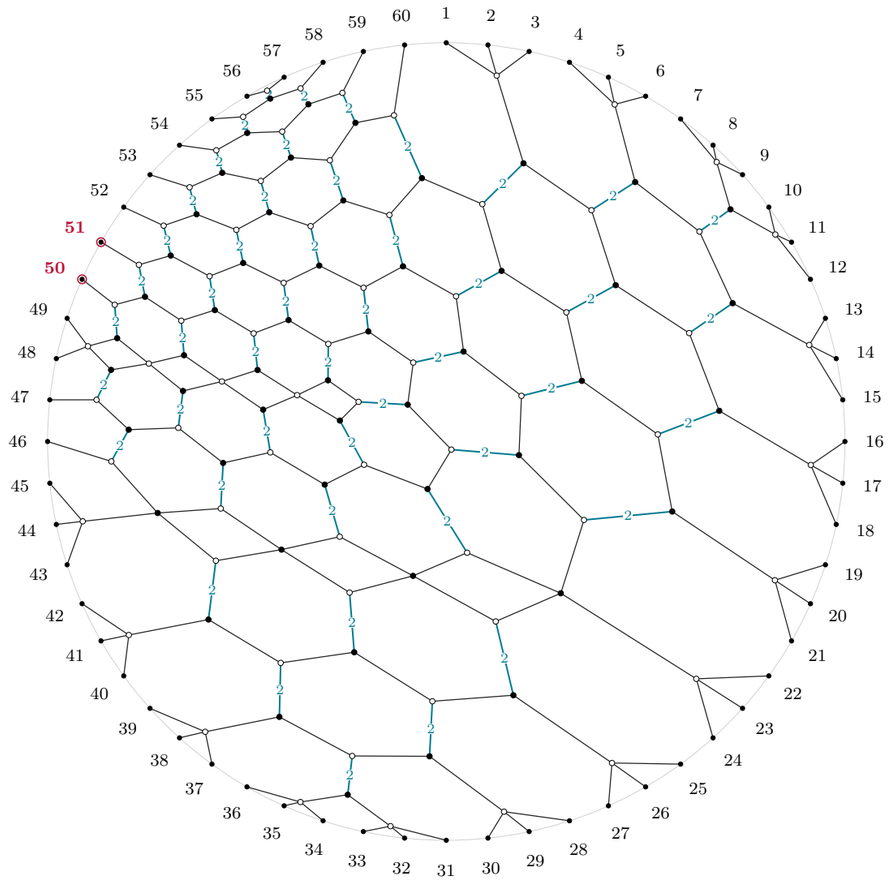


Tableau

1	4	7	10	13	16	19	22	25	28	31	40	43	47	48
2	5	8	11	14	20	26	29	32	34	37	44	50	52	54
3	6	9	17	23	27	33	35	38	41	46	51	53	55	56
12	15	18	21	24	30	36	39	42	45	49	57	58	59	60

Target: $\ell = 943$

119 internal vertices; 48 double edges



Tableau

1	4	7	10	13	16	19	22	25	28	31	34	37	40	43
2	5	8	11	14	17	20	23	26	29	32	44	46	47	48
3	6	9	21	24	27	30	33	35	38	41	53	54	55	56
12	15	18	36	39	42	45	49	50	51	52	57	58	59	60

Red positions: $50, 51$; source colors BC, target colors DD.

Teal edges labelled 2 denote double hourglasses. The boundary circle is a drawing guide.

Growth representatives for the canonical labels; not asserted equal to the canonical tensors.

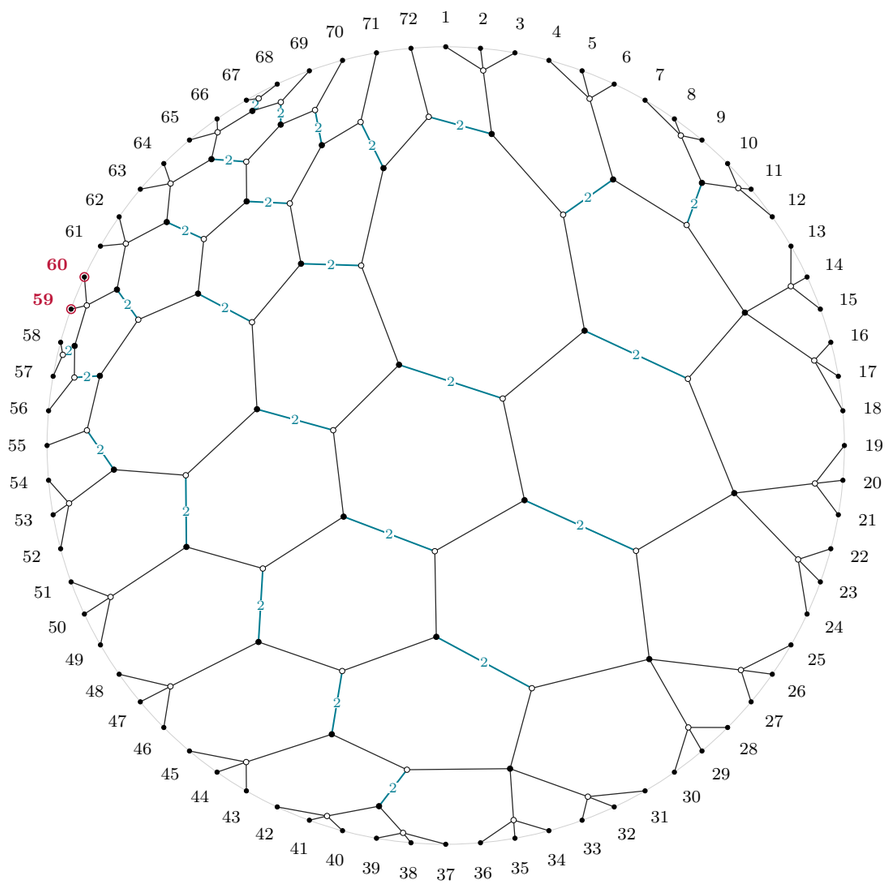
The coefficient is supplied by the transfer argument, not evaluated from these diagrams.

Uniform four-row labels: $m = 5$, shape $(18)^4$

Transfer formula: $\mu = 6$ S_{72} Length difference 129 Critical q-degree 64

Source: $\ell = 1231$

78 internal vertices; 26 double edges

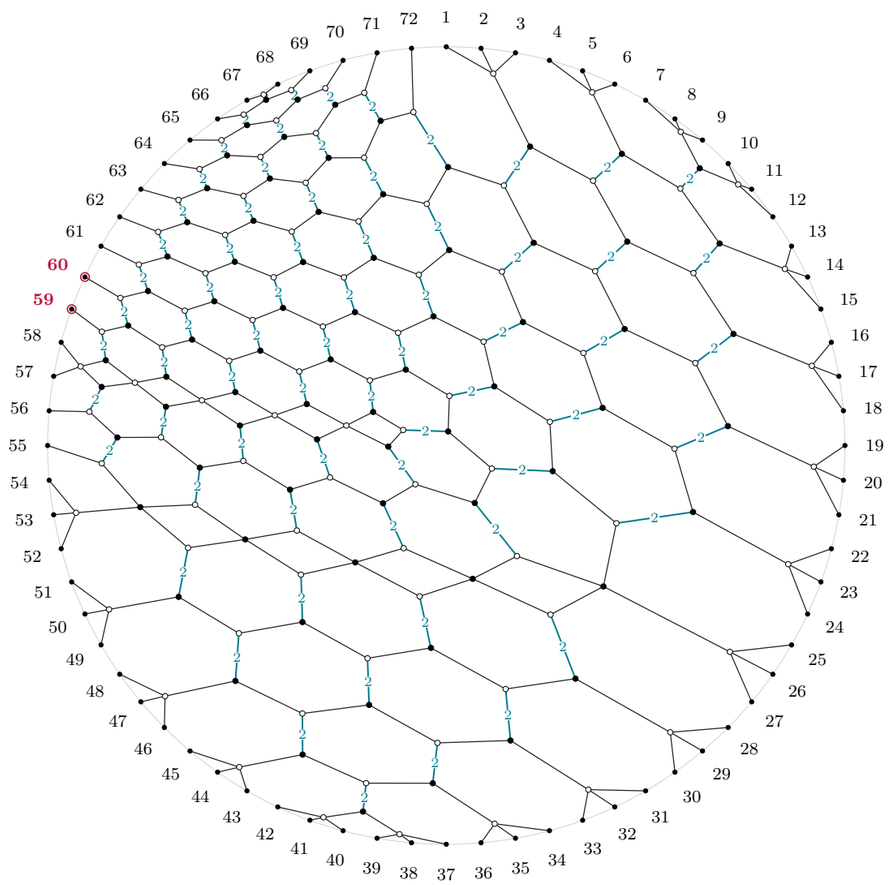


Tableau

1	4	7	10	13	16	19	22	25	28	31	34	37	46	49	52	56	57
2	5	8	11	14	20	26	29	32	35	38	40	43	53	59	61	63	65
3	6	9	17	23	27	33	39	41	44	47	50	55	60	62	64	66	67
12	15	18	21	24	30	36	42	45	48	51	54	58	68	69	70	71	72

Target: $\ell = 1360$

168 internal vertices; 70 double edges



Tableau

1	4	7	10	13	16	19	22	25	28	31	34	37	40	43	46	49	52
2	5	8	11	14	17	20	23	26	29	32	35	38	50	53	55	56	57
3	6	9	21	24	27	30	33	36	39	41	44	47	63	64	65	66	67
12	15	18	42	45	48	51	54	58	59	60	61	62	68	69	70	71	72

Red positions: $59, 60$; source colors BC, target colors DD.

Teal edges labelled 2 denote double hourglasses. The boundary circle is a drawing guide.

Growth representatives for the canonical labels; not asserted equal to the canonical tensors.

The coefficient is supplied by the transfer argument, not evaluated from these diagrams.